

SMART STORAGE MONITOR SYSTEM



CSE 326: Micro Processor & Micro Controller Lab Project

SUBMITTED BY

Name	ID	Intake
Al Adid	20245103007	53
Md Fahim Islam	20245103035	53
Md Mahfuzul Bashir Shaikot	20245103038	53
Md. Sakibur Rahman	20245103091	53
Abu Ahmad Mahfuzor Rahman	20234103092	53

SUPERVISED BY:

Fahima Khanam
*Assistant Professor,
Department of CSE*

**DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
BANGLADESH UNIVERSITY OF BUSINESS AND TECHNOLOGY
(BUBT)**

16, July, 2026

Project Name : Smart Storage Monitor System

Objective :

Smart Storage Monitor System is a security system developed using Arduino Uno technology for the purpose of constantly monitoring environmental conditions in storage rooms, warehouses, and storage spaces. The system is aimed at ensuring safety of goods stored in these environments against destruction and any form of theft.

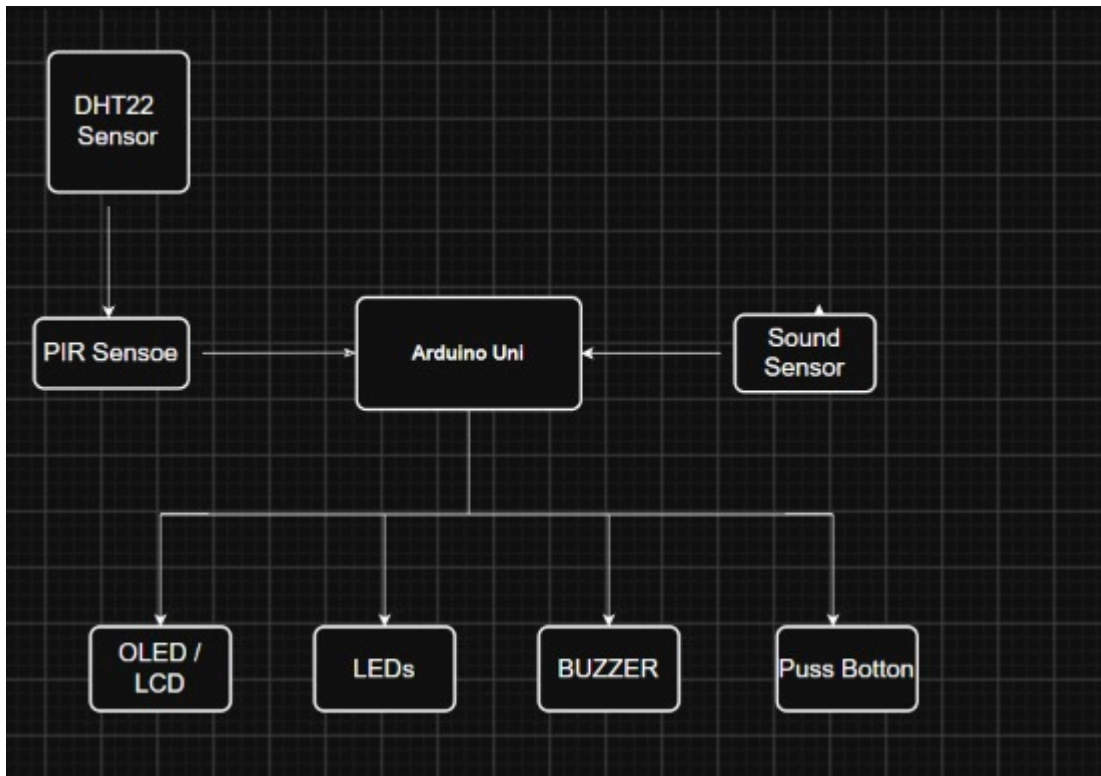
Functions :

- Sense temperature in real time with the use of the DHT22 sensor.
- Sensing humidity in real time with the use of the DHT22 sensor.
- Movement sensing with the PIR motion sensor.
- Sound sensing for any abnormal sound detection.
- Visual representation of temperature and humidity on an OLED/LCD display.
- Temperature and humidity status displayed by symbols such as ↑ (High), ↓ (Low), and ✓ (Normal).
- Generate alerts via LEDs.
- Sound alert via buzzer when any abnormal situation is sensed.
- Reset/Arming feature via push button.

Required equipment (hardware) :

- Arduino Uno
- DHT22 Temperature & Humidity Sensor
- PIR Motion Sensor (HC-SR501)
- Sound Sensor (KY-037/KY-038)
- 0.96" OLED Display / 16×2 I2C LCD
- Red LED (Temperature Status)
- Blue LED (Humidity Status)
- Yellow LED (Motion Status)
- White/Purple LED (Sound Status)
- Green LED (System Status)
- Active Buzzer
- Push Button (Reset/Arm)
- Breadboard or PCB
- 220Ω Resistors
- Jumper Wires As Required
- 5V Power Supply / USB Adapter

Project Design :



Working Principle :

- DHT22 sensor continuously monitors temperature and humidity levels.
- PIR sensor senses any motion detected within the storage area.
- Sound sensor senses any abnormal sound level.
- Data from all sensors is processed by Arduino Uno.
- Current values and status are shown on the OLED/LCD display.
- LEDs show the status of temperature, humidity, motion, sound, and the system as a whole.
- In case any value crosses the defined threshold, then buzzer sounds an alarm.
- Push button resets the alarm or arming of the system.

Project Application :

- Warehouses and storage rooms
- Food storage rooms
- Storage rooms for pharmaceuticals
- Library archive rooms

- Data centers and server rooms
- Home storage rooms
- Record rooms in offices
- Inventory control in industries
- Advantages
- It protects the items stored from any damage due to environmental factors.
- Improves security of the storage room with the help of motion and sound sensors.
- Real-time monitoring and alerts facility.
- Manual efforts saved.
- Affordable and easy-to-install system.